Lecture 7: By Force of Habit

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Intro

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- Late 80s: everyone started for a solution to the equity premium puzzle
- Many solutions proposed: more complicated preferences, market frictions, aggregation issues...
- There is no single solution – many mechanisms are at play
- No notion of "better" or "worse"
- Everyone was looking for the same: how can we *spice up* the SDF and get more volatility?
- The next few classes will be about "wins" in this game

Flight Plan

- Parts I-II: derive the Campbell-Cochrane habit model (Campbell and Cochrane 1999)
- Part III: the model meets (and matches!) the data
- Part IV: follow-up tracks in equity markets, fixed income securities, and foreign exchange (FX)

Part I: By Force of Habit – Main Setup

Why Habit?

- Time-separable CRRA struggles with high equity premia and low/steady r_f (Mehra and Prescott 1985; Weil 1989)
- We want countercyclical risk prices without implausible interest-rate dynamics
- External habit creates a recession state variable
- Target: high Sharpe ratios in bad times

Preferences and Surplus Consumption

- Representative agent has utility on consumption levels *net* of habit:

$$U_t = \mathbb{E}_t \left[\sum_{j=0}^{\infty} \beta^j \frac{(C_{t+j} - X_{t+j})^{1-\gamma} - 1}{1-\gamma} \right]$$

- X_t: habit, C_t: consumption, β: discount factor, γ: power parameter
- Habit is *external*: your individual consumption does not affect X_t
- "Keeping up with the Joneses": utility depends on relative consumption
- *Internal* habit: X_t depends on your own past consumption, not aggregate
- External habit is way more tractable!

State-Dependent Risk Aversion

- Define the surplus consumption ratio and its log state:

$$S_t \equiv \frac{C_t - X_t}{C_t}, \qquad s_t \equiv \log S_t$$

- Local curvature is state dependent:

$$-\frac{C \cdot u''(C-X)}{u'(C-X)} = \gamma \frac{C}{C-X} = \frac{\gamma}{S}$$

- Low $S_t \implies$ high local curvature
- As S_t moves around, risk-aversion moves around

Consumption and Habit Dynamics

- Consumption growth is i.i.d. lognormal ($c_t \equiv \log(C_t)$):

$$\Delta c_{t+1} = g + \sigma_c \varepsilon_{c,t+1}, \qquad \varepsilon_{c,t+1} \sim \mathcal{N}(0,1)$$

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- **Exogenous** law of motion for s_t:

$$s_{t+1} = (1-\phi)\bar{s} + \phi s_t + \lambda(s_t)\sigma_c \varepsilon_{c,t+1}$$

- ϕ: persistence, $\bar{s}$: mean surplus consumption, g: mean consumption growth, $\lambda(s_t)$: sensitivity function

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- Exercise 1: assume that $\sigma_c = 0$ and use a Taylor expansion to show that, for some α

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- Exercise 2: (using the result above) show that $x_{t+1} = \alpha + (1-\phi)\sum_{j=0}^{\infty} \phi^j c_{t-j}$

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$$M_{t+1} \equiv \beta \frac{u'(C_{t+1} - X_{t+1})}{u'(C_t - X_t)} = \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} \left(\frac{S_{t+1}}{S_t} \right)^{-\gamma}$$

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- The log-SDF m_{t+1} follows the following process:

$$m_{t+1} = \log(\beta) - \gamma g + \gamma(1 - \phi)(s_t - \bar{s}) - \gamma \sigma_c [1 + \lambda(s_t)] \varepsilon_{c,t+1}$$

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- The risk-free rate is $r_{f,t} = \log(\mathbb{E}_t[M_{t+1}])$ is available in closed form:

$$\boxed{r_{f,t} = -\log(\beta) + \gamma g - \gamma(1-\phi)(s_t - \bar{s}) - \frac{\gamma^2 \sigma^2}{2} [1 + \lambda(s_t)]^2}$$

- Exercise 3: show it at home
- What's the intuition for each term?

Deriving the Sharpe Ratio Bound

- From the log-SDF: $m_{t+1} = \log(\beta) - \gamma g + \gamma(1-\phi)(s_t - \bar{s}) - \gamma\sigma_c[1 + \lambda(s_t)]\varepsilon_{c,t+1}$
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- From the log-SDF: $\sigma_t(m_{t+1}) = \gamma\sigma_c[1 + \lambda(s_t)]$
- Therefore: $\max_R \frac{\mathbb{E}_t[R^e_{t+1}]}{\sigma_t(R^e_{t+1})} = \sqrt{\exp(\gamma^2\sigma_c^2[1 + \lambda(s_t)]^2) - 1} \approx \gamma\sigma_c[1 + \lambda(s_t)]$
- The Sharpe ratio bound will move around with s_t through $\lambda(s_t)$

Part II: By Force of Habit – Calibrating $\lambda(s_t)$

How to choose $\lambda(s_t)$?

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Let's match the following features:

1. The (real) risk-free rate should be constant
2. Today's consumption does not affect today's habit if $s_{t+1} \approx \bar{s}$: $(dx_{t+1}/dc_{t+1} = 0)$
3. For any value of s_t, the habit response to consumption shocks should be nonnegative: $(dx_{t+1}/dc_{t+1} \geq 0)$

Restricting $\lambda(s_t)$

- Functional-form choice in Campbell and Cochrane (1999):

$$\lambda(s_t) \equiv \begin{cases} \frac{1}{\bar{S}}\sqrt{1-2(s_t-\bar{s})}-1, & s_t \leq s_{\max} \\ 0, & s_t > s_{\max} \end{cases}, \quad s_{\max} \equiv \bar{s} - \frac{1}{2}(1-\bar{S}^2),$$

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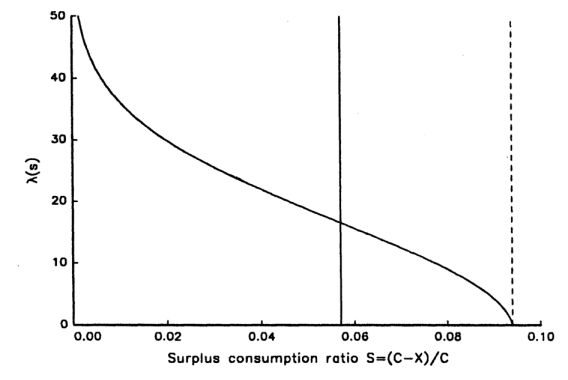
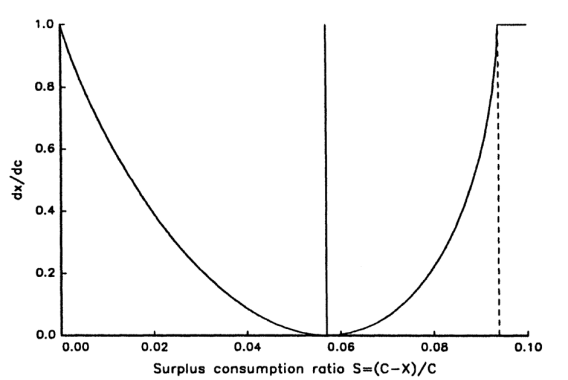
- It is *decreasing* with s_t: habit becomes less sensitive to shocks as s_t increases
- Exercise 4: show that a function $\lambda(s_t)$ can make $r_{f,t}$ constant if, and only if

$$(1+\lambda(s_t)) \cdot \lambda'(s_t) = \frac{1-\phi}{\gamma\sigma_c} = \frac{1}{\bar{S}^2}$$

How This Function Looks Like





- Solid line: steady-state value $\bar{S}$
- Dashed line: maximum value $S_{\max}$, above which $\lambda(s_t) = 0$
- Important: dx/dc is U-shaped, with a minimum at $\bar{S}$ where it is zero

Does It Do Its Job?

Constant risk-free rate:

$$r_{f,t} = -\log(\beta) + \gamma g - \gamma(1-\phi)(s_t - \bar{s}) - \frac{\gamma^2\sigma^2}{2}[1+\lambda(s_t)]^2 = -\log(\beta) + \gamma g - \frac{\gamma}{2}(1-\phi)$$

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Habit response to consumption:

$$\frac{dx_{t+1}}{dc_{t+1}}(s_{t+1}) = 1 + \frac{\lambda(s_{t+1})}{1-e^{-s_{t+1}}} = 1 + \frac{\frac{1}{S}\sqrt{1-2(s_{t+1}-\bar{s})}-1}{1-e^{-s_{t+1}}} = 1 + \frac{\lambda(s_{t+1})}{1-e^{-s_{t+1}}}$$

$$\frac{dx_{t+1}}{dc_{t+1}}(\bar{s}) = 1 + \frac{\frac{1}{S}-1}{1-\frac{1}{S}} = 0$$

- Exercise 5: show that $dx/dc \geq 0$ for all s_{t+1}

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$$P_t^c = \mathbb{E}_t\left[M_{t+1}\left(P_{t+1}^c + C_{t+1}\right)\right]$$

Divide by C_t:

$$\boxed{V(s_t) = \mathbb{E}\left[M_{t+1}\frac{C_{t+1}}{C_t}\left(1 + V(s_{t+1})\right) \mid s_t\right]}$$

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- In Campbell–Cochrane:

$$\frac{C_{t+1}}{C_t} = e^{g+\sigma_c\varepsilon_t}, \qquad s_{t+1} = (1-\phi)\bar{s} + \phi s_t + \lambda(s_t)\sigma_c\varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0,1)$$

The One-Shock Integrand

Recall:

$$M_{t+1} = \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} \left(\frac{S_{t+1}}{S_t} \right)^{-\gamma} = \beta \left(e^{g+\sigma_c \varepsilon_t} \right)^{-\gamma} e^{-\gamma(s_{t+1}-s_t)}$$

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With one shock ε, you need to approximate an integral over $\mathbb{R}$:

$$V(s) = \int_{-\infty}^{\infty} \underbrace{\beta e^{(1-\gamma)(g+\sigma_c \varepsilon)} e^{-\gamma(s'(\varepsilon;s)-s)} \left(1 + V(s'(\varepsilon;s)) \right)}_{\text{known given } V(\cdot)} \varphi(\varepsilon)\, d\varepsilon$$

where $\varphi(\varepsilon)$ is the standard normal density and $s'(\varepsilon;s) = (1-\phi)\bar{s} + \phi s + \lambda(s)\sigma_c \varepsilon$

Pricing a Dividend Claim

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- Their idea: $\Delta d_t = g + \sigma_w w_t$, where corr$(w_t, \varepsilon_t) = 0.2$
- Define the price-divided ratio as $Q(s_t) \equiv \frac{P_t^d}{D_t}(s_t)$. It will follow:

$$Q(s_t) = \mathbb{E}\left[M_{t+1}\frac{D_{t+1}}{D_t}\left(1 + Q(s_{t+1})\right) \mid s_t\right]$$

- You can solve this with the same type of technique as before
- Notice that the randomness on M_{t+1} is only through ε_t

Questions?

Part III: Meet The Data

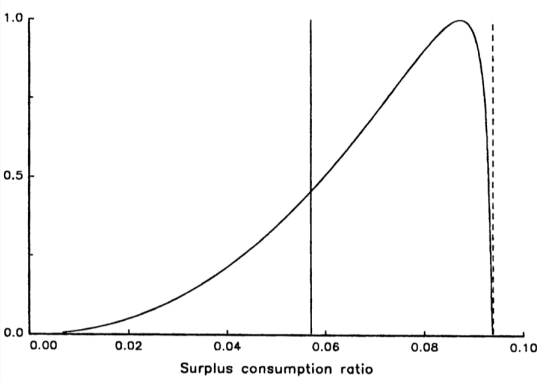
Calibration

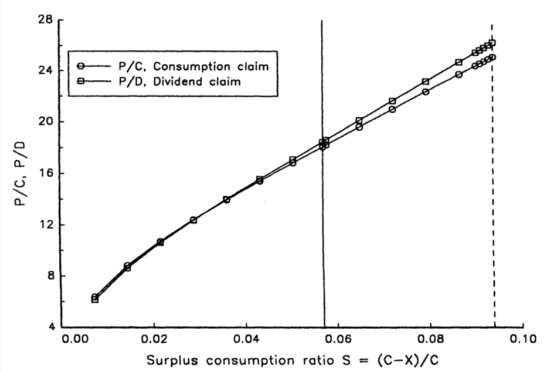
PARAMETER CHOICES

Parameter	Variable	Value
Assumed:		
Mean consumption growth (%)*	g	1.89
Standard deviation of consumption growth (%)*	σ	1.50
Log risk-free rate (%)*	r^f	.94
Persistence coefficient*	ϕ	.87
Utility curvature	γ	2.00
Standard deviation of dividend growth (%)*	σ_w	11.2
Correlation between Δd and Δc	ρ	.2
Implied:		
Subjective discount factor*	δ	.89
Steady-state surplus consumption ratio	$\overline{S}$	.057
Maximum surplus consumption ratio	$S_{\max}$	.094

* Annualized values, e.g., $12g$, $\sqrt{12}\sigma$, $12r^f$, ϕ^{12}, and δ^{12}, since the model is simulated at a monthly frequency.

Surplus Consumption and Asset Prices





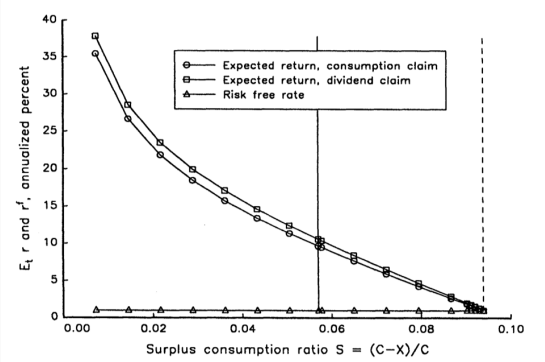
Equity Premium Puzzle No More!

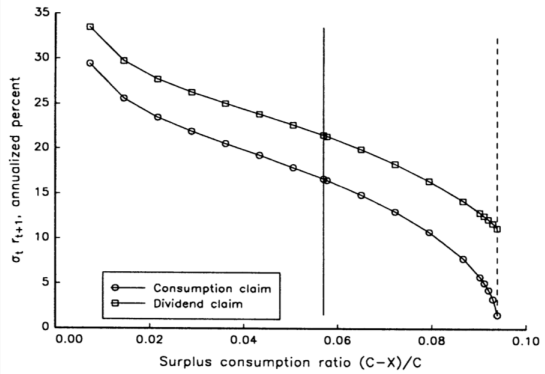
TABLE 2

MEANS AND STANDARD DEVIATIONS OF SIMULATED AND HISTORICAL DATA

Statistic	Consumption Claim	Dividend Claim	Postwar Sample	Long Sample
$E(\Delta c)$	1.89*		1.89	1.72
$\sigma(\Delta c)$	1.22*		1.22	3.32
$E(r^f)$	.094*		.094	2.92
$E(r - r^f)/\sigma(r - r^f)$	.43*	.33	.43	.22
$E(R - R^f)/\sigma(R - R^f)$	.50		.50	
$E(r - r^f)$	6.64	6.52	6.69	3.90
$\sigma(r - r^f)$	15.2	20.0	15.7	18.0
$\exp[E(p - d)]$	18.3	18.7	24.7	21.1
$\sigma(p - d)$	.27	.29	.26	.27

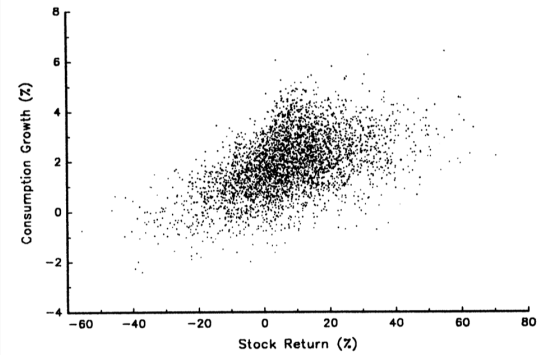
Equity Premium and Conditional Volatility

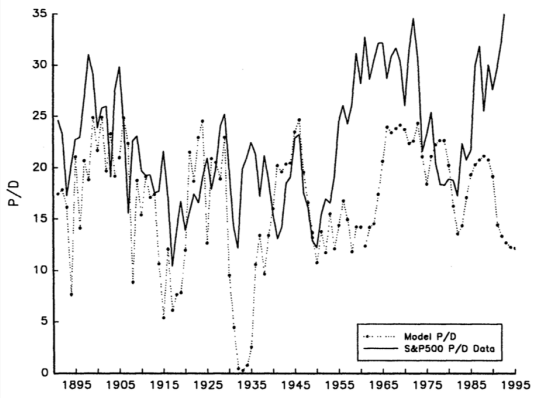




- The equity premium is large *and* it is time-varying!
- Return volatility is high and countercyclical, even with smooth consumption growth

Consumption Growth, Stock Returns, and P/D Ratio





Questions?

Part IV: Where Habit Models Went

Three Follow-Up Tracks

At least three quite cool direct follow-ups:

- More on equity: richer predictability and valuation implications (Menzly, Santos, and Veronesi 2004)
- Interest rate term structure: bond premia under habit-based discount rates (Wachter 2006)
- FX markets: currency premia from cross-country habit risk (Verdelhan 2010)
- Same mechanism: state-dependent SDF via S_t
- Exercise 6: read these papers!

Questions?

Takeaways

- External habit resolves key aggregate puzzles through state-dependent risk prices
- Main trick: time-varying risk-aversion even with smooth consumption growth
- Calibration discipline is essential: $\lambda(s_t)$ is doing the hard work
- Pushback: is this truly microfounded?
- Follow-ups show the same mechanism in equities, bonds, and FX

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- External habit resolves key aggregate puzzles through state-dependent risk prices
- Main trick: time-varying risk-aversion even with smooth consumption growth
- Calibration discipline is essential: $\lambda(s_t)$ is doing the hard work
- Pushback: is this truly microfounded?
- Follow-ups show the same mechanism in equities, bonds, and FX
- Next class: time-varying disaster risk!

Readings and References i

- Campbell, Chapter 6

Campbell, John Y., and John H. Cochrane. 1999. "By Force of Habit: A Consumption-Based Explanation of Aggregate Stock Market Behavior." *Journal of Political Economy* 107 (2): 205–51. https://doi.org/10.1086/250059.

Mehra, Rajnish, and Edward C. Prescott. 1985. "The Equity Premium: A Puzzle." *Journal of Monetary Economics* 15 (2): 145–61. https://doi.org/10.1016/0304-3932(85)90061-3.

Menzly, Lior, Tano Santos, and Pietro Veronesi. 2004. "Understanding Predictability." *Journal of Political Economy* 112 (1): 1–47. https://doi.org/10.1086/379934.

Verdelhan, Adrien. 2010. "A Habit-Based Explanation of the Exchange Rate Risk Premium." *The Journal of Finance* 65 (1): 123–46. https://doi.org/10.1111/j.1540-6261.2009.01525.x.

Wachter, Jessica A. 2006. "A Consumption-Based Model of the Term Structure of Interest Rates." *Journal of Financial Economics* 79 (2): 365–99. https://doi.org/10.1016/j.jfineco.2005.02.004.

Readings and References ii

Weil, Philippe. 1989. "The Equity Premium Puzzle and the Risk-Free Rate Puzzle." *Journal of Monetary Economics* 24 (3): 401–21. https://doi.org/10.1016/0304-3932(89)90028-7.